

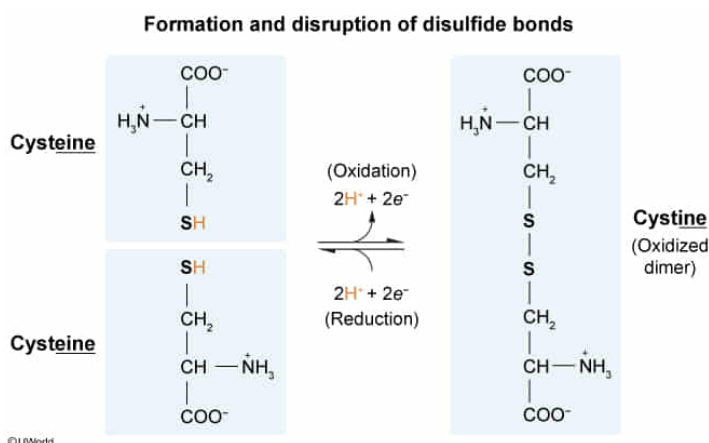
Which statement correctly describes the cleavage of protein disulfide bonds by the enzyme thioredoxin?

- ☐ A. A protease breaks the bond by catalyzing the addition of water across it. [21%]
- ☐ B. An isomerase catalyzes a rearrangement of functional groups to break the bond. [2%]
- ☐ C. A phosphorylase breaks the bond by catalyzing the addition of a phosphate group. [3%]
- ✓ ☒ D. An oxidoreductase catalyzes the transfer of electrons to break the bond. [72%]

Correct

72% answered correctly

## Explanation:



Oxidation is the loss of electrons that occurs when an electron donor transfers electrons to an acceptor. The reverse process, referred to as a reduction, involves the gain of electrons. In general, reactions in which an acceptor becomes reduced and a donor becomes oxidized are called **oxidation-reduction**, or **redox**, reactions. Enzymes that catalyze redox reactions are collectively known as **oxidoreductases**.

In proteins, **disulfide bonds** form when the thiol groups of **cysteine** residues are **oxidized**, linking them together to form **cystine**. To break the disulfide bond in cystine, a reduction step that transfers electrons back to the sulfur atoms is required. **Thioredoxin catalyzes the**

**transfer of electrons** from a donor molecule to cystine, which reduces the sulfur atoms and oxidizes the donor molecule. Consequently, thioredoxin can be classified as an oxidoreductase because it facilitates the cleavage (reduction) of disulfide bonds.

**(Choice A)** Proteases are a type of **hydrolase** that **break peptide bonds** by hydrolysis (addition of water). They do not break disulfide bonds.

**(Choice B)** **Isomerases** rearrange functional groups within a molecule but do not catalyze redox reactions. Therefore, they cannot catalyze cleavage of disulfide bonds.

**(Choice C)** Phosphorylases are a type of **transferase** that break bonds by adding phosphate, as in **glycogen degradation**. They do not catalyze redox reactions and cannot break disulfide bonds.

### **Educational objective:**

Disulfide bonds form in proteins when two cysteine residues are oxidized (lose electrons) to form cystine. They must be reduced (gain electrons) for the bond to break. Enzymes that catalyze oxidation-reduction reactions (transfer of electrons) are collectively known as oxidoreductases.

**Time spent:**0 sec

**Subject:**  
Biochemistry

**QID:**402110

**System:**  
Enzymes